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**TITLE: An Integrated Geo Intelligence Framework
Linking Site Suitability, Spatial Customer Segmentation
and Delivery Planning for Urban Restaurant Location
Decisions: A Case Study of Delhi, India**

Abstract

Restaurant location decisions in dense Indian cities are typically resolved through local knowledge and brokerled judgement, and the three questions that determine commercial viability — where to locate, whom to serve, and how to deliver — are usually answered separately by different actors using different evidence. This study asks whether these three decisions can be addressed within a single, transparent, reproducible geospatial framework. A modular GeoAI system was developed for the National Capital Territory of Delhi, integrating three components: a multi-criteria weighted-overlay suitability model applied to 5,946 candidate locations generated on a 500 m grid and characterised using OpenStreetMap road-network and point-of-interest features; density-based spatial clustering (DBSCAN) of customer order records to delineate demand segments; and a nearest-neighbour with 2-opt improvement heuristic for delivery-route sequencing from the highest-ranked site. The suitability surface, classified into five bands from Very Low (0.20-0.31) to Very High (0.66-0.77), formed discontinuous corridors and nodes rather than a single contiguous zone; a discrete candidate-grid ranking identified a leading site combining 34.3 m proximity to a main road with two malls and two metro stations within 500 m. Clustering resolved five spatially distinct customer segments, with 12 per cent of customers retained as unclustered noise under DBSCAN and further mapped by indicative value tier using k-means. Because the transaction layer is synthetic, findings establish methodological feasibility rather than validated commercial demand. The principal contribution is architectural: demonstrating that strategic, tactical and operational location decisions can be linked through auditable spatial evidence within one workflow.

Keywords: Geo Intelligence, restaurant site selection, multi-criteria decision analysis, spatial customer segmentation, last-mile delivery optimisation

Introduction

The location of a restaurant is among the most consequential and least reversible decisions in food-service enterprise planning. It simultaneously governs visibility, walk-in footfall, exposure to competing establishments, accessibility for staff and customers, and the practical geography within which delivery orders must be fulfilled. In dense metropolitan settings these considerations frequently conflict: the most accessible commercial nodes are precisely those where competitive saturation is highest, while districts generating substantial order value may lie at distances that erode delivery

margins. Conventional practice in Indian cities addresses these tensions through accumulated local knowledge and broker-mediated judgement, an approach that is neither systematically auditable nor transferable between decision-makers.

Geospatial artificial intelligence offers a structured alternative. GeoAI has been characterised as the application of spatially explicit artificial intelligence techniques to geographic knowledge discovery and decision support (Janowicz et al., 2020), and subsequent work has emphasised the importance of transparency and explainability where such systems inform consequential decisions (Li et al., 2024). Applied to restaurant planning, this framing permits road networks, amenity distributions, competitor locations and customer transactions to be converted into comparable spatial evidence rather than remaining as disconnected qualitative impressions.

The research problem addressed here is not the absence of analytical methods for any individual component. Multi-criteria site suitability analysis, density-based spatial clustering and vehicle-routing optimisation are each supported by mature and well-validated literatures. The problem is that these methods are conventionally deployed in isolation, producing evidence that answers one question while remaining silent on the others. A site may score highly on accessibility criteria while lying outside the delivery reach of the customer concentrations that would sustain it; conversely, a valuable demand cluster is of limited operational relevance if it cannot be served reliably. The decisions are analytically separable but commercially interdependent, and existing applied work rarely reflects that interdependence — particularly for Delhi, where peer-reviewed evidence applying these methods jointly is sparse.

This study therefore develops and demonstrates an integrated three-module GeoAI framework for the National Capital Territory of Delhi, using restaurants as the application domain. The objectives are to construct a transparent multi-criteria suitability model capable of ranking candidate locations across the study area using openly available spatial data; to characterise the geographic structure of customer demand through spatial clustering and to profile the resulting segments by order volume and value; to translate a selected location into operational delivery geography by defining service catchments and sequencing an optimised delivery route; and, critically, to link these three components such that the outputs of the site-selection and customer-analysis modules become the direct inputs to the delivery-planning module. The intended contribution is architectural rather than algorithmic: no new estimator is proposed, and the value claimed lies in demonstrating that strategic, tactical and operational location decisions can be connected through visible, auditable spatial evidence within a single reproducible workflow.

Related Work

The methodological foundations of this study draw on four distinct research streams, each mature in isolation but rarely combined in applied urban settings.

Work on Geo Intelligence as a coherent field provides the framing for the

present study. Janowicz et al. (2020) established the scope of spatially explicit artificial intelligence for geographic knowledge discovery, arguing that spatial structure must be treated as analytically central rather than as incidental metadata. Mai et al. (2022) subsequently reviewed location-encoding strategies through which spatial context is represented computationally, while Li et al. (2024) emphasised responsible and explainable GeoAI, noting that systems informing consequential decisions must expose their criteria to inspection. This last consideration directly motivated the present study's preference for a transparent weighted-overlay formulation over a supervised learning model whose internal weighting would be less readily interpretable to non-technical stakeholders.

Retail site selection has a substantial GIS heritage grounded in multi-criteria decision analysis. Roig-Tierno et al. (2013) combined geographic information systems with the analytic hierarchy process to support supermarket site selection, demonstrating that spatial standardisation and explicit criterion weighting can be integrated within a single decision framework. The underlying logic — that location suitability is inherently multi-criteria, since accessibility, demand generation and competitive pressure typically favour different sites — is adopted directly in the present work, with road density, main-road proximity, competitor context, transit access and commercial amenity presence treated as separate weighted criteria rather than collapsed into a single composite proxy.

The reliance of such analyses on volunteered geographic information introduces well-documented data-quality considerations. Haklay (2010) established the foundational comparative assessment of OpenStreetMap against authoritative survey data, quantifying both positional accuracy and, more consequentially for amenity-based analysis, completeness variability across areas. Klinkhardt et al. (2023) extended this specifically to points of interest, benchmarking OSM POI completeness against large-scale reference data and confirming that amenity coverage should be treated as a sample of mapped activity rather than as an exhaustive establishment census. This caution proved directly material to the present study, where certain amenity categories were found to be substantially under-represented in the retrieved data.

Spatial clustering for market segmentation rests on the density-based approach introduced by Ester et al. (1996), whose DBSCAN algorithm identifies arbitrarily shaped clusters in spatial databases while explicitly labelling low-density observations as noise rather than forcing their assignment to a nearest centroid. This property is the principal reason DBSCAN was preferred here over partitional alternatives: urban customer distributions characteristically combine dense commercial concentrations with dispersed peripheral activity, and compelling the latter into clusters distorts segment profiles. Schubert et al. (2017) subsequently clarified parameterselection practice and reaffirmed the algorithm's continued suitability for spatial applications, while noting that reported results should always be accompanied by the neighbourhood-radius and minimum-points values used.

Finally, network-based accessibility and routing analysis has been made

substantially more tractable by Boeing

(2017), whose OSMnx framework enables street networks to be retrieved, constructed and analysed programmatically. Service-area delineation and route sequencing in the present study follow from this capability, although the delivery catchments reported here remain Euclidean approximations rather than full network travel-time isochrones — a simplification acknowledged explicitly in the discussion.

Considered together, these literatures supply robust treatments of each individual component while leaving the integration question largely unaddressed. The present study occupies that gap.

Methods

Study area

The study covers the National Capital Territory of Delhi, extending approximately from 28.41° N to 28.88° N and from 76.85° E to 77.35° E. Delhi presents a heterogeneous commercial landscape encompassing established central business districts, planned sub-city developments and peripheral residential zones, with correspondingly variable transport connectivity and metro accessibility. This internal diversity makes it a suitable setting for demonstrating a location-intelligence framework whose value lies precisely in discriminating between spatially adjacent but functionally distinct opportunities.

Analytical framework

The system comprises three sequentially linked modules, the architecture of which is shown in Figure 1. Module 1 screens the urban context to identify and rank candidate locations. Module 2 characterises the geography and value structure of customer demand. Module 3 converts a selected location into operational delivery geography. Integration is achieved by construction rather than by interpretation: the restaurant location used in Module 3 is drawn directly from the highest-ranked output of Module 1, and the delivery destinations are sampled from the customer records analysed in Module 2.

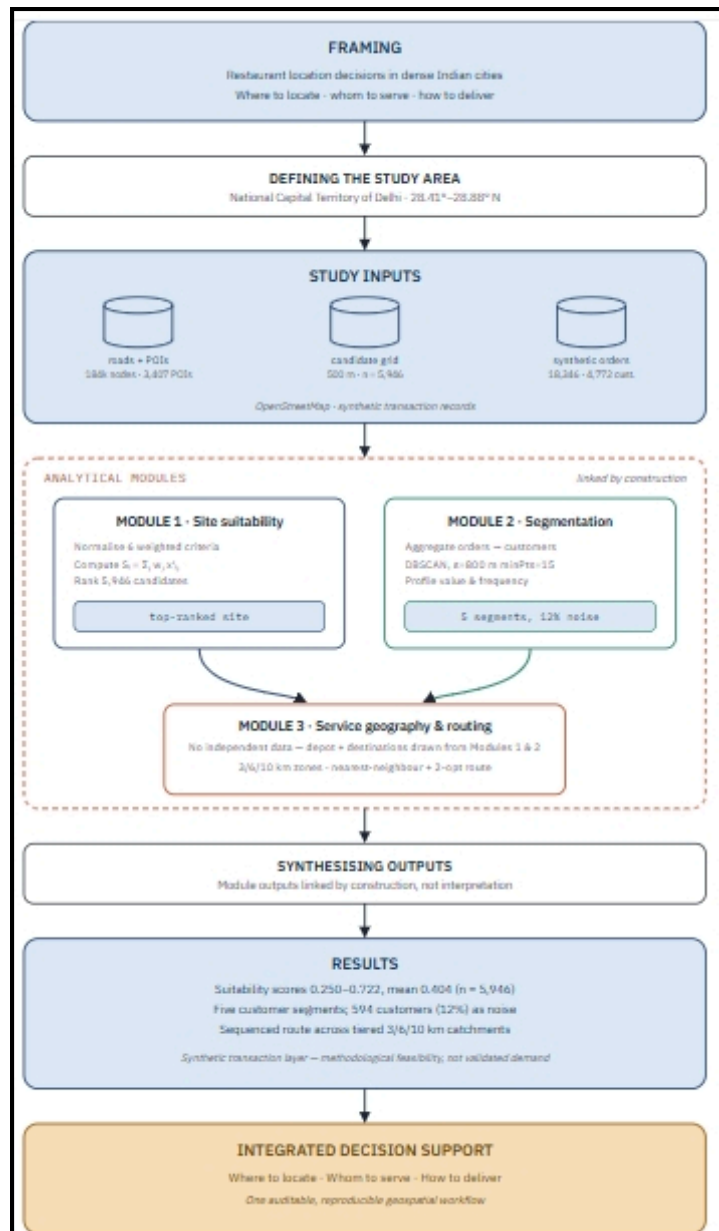


Figure 1. Integrated analytical framework showing the three modules and their linkage into a combined decision-support output.

Data and preprocessing

The road network and point-of-interest layers were obtained from OpenStreetMap. The retrieved drivable network comprised 184,036 nodes and 499,988 edges. Point-of-interest extraction returned 3,407 records, of which 3,062 were classified as competing food-service establishments (restaurants, cafés and fast-food outlets), 244 as metro stations, 51 as colleges or universities and 28 as shopping malls.

All layers were harmonised to a common coordinate reference framework. Data were stored and displayed in geographic coordinates (EPSG:4326) but

reprojected to Universal Transverse Mercator Zone 43N

(EPSG:32643) for all distance, buffer and area computations, ensuring that metric operations were performed in linear units rather than in angular degrees. This distinction is consequential: distance calculations executed directly in geographic coordinates yield systematically incorrect results without raising computational errors.

Customer transaction data were not available from commercial sources owing to privacy constraints. A synthetic order dataset was therefore generated comprising 18,346 orders attributed to 4,772 unique customers over a three-month period, with customer locations distributed around identifiable Delhi neighbourhoods and order values, cuisine categories and temporal patterns assigned to produce plausible structure. This dataset supports methodological demonstration only; no inference regarding actual consumer behaviour is drawn from it, and this constraint is carried explicitly through the results and discussion.

Module 1: multi-criteria site suitability

A regular grid of candidate locations was generated at 500 m spacing across the study area and clipped to the administrative boundary, yielding 5,946 evaluable points. For each candidate, six criteria were computed within a 500 m analysis neighbourhood: road density, expressed as total mapped road length; distance to the nearest primary, secondary or trunk road; competitor count and distance to nearest competitor; count of offices and educational institutions; count of shopping malls; and count of metro stations.

Criteria were standardised by min-max normalisation to a common zero-to-one range. Cost criteria, for which lower values are favourable, were inverted following normalisation so that higher standardised values consistently denote greater favourability. The composite suitability score for candidate i was computed as the weighted sum

$$S_i = \sum_j w_j x'_{ij}, \text{ where } \sum_j w_j = 1$$

with x'_{ij} denoting the standardised value of criterion j at candidate i and w_j its assigned weight. Weights were assigned on reasoned grounds reflecting the relative importance of accessibility, demand generation and competitive pressure. It is acknowledged that these were not derived through formal stakeholder elicitation or a documented analytic hierarchy process, and that sensitivity analysis over the weight vector represents a necessary extension.

For cartographic presentation, the composite score was additionally rendered as a continuous 100 m raster across the full study area in Google Earth Engine, combining gridded population density (WorldPop), road density, distance to the nearest metro station and competitor density, with final map production completed in ArcGIS Pro. This raster (Figure 2) provides complete spatial coverage for visual interpretation, complementing the discrete candidate-grid ranking (Table 1) used for shortlisting.

Module 2: spatial customer segmentation

Order records were aggregated to customer level, yielding for each customer a representative location, total order count, aggregate expenditure and mean order value. Customer locations were projected to EPSG:32643 and clustered using DBSCAN with a neighbourhood radius of 800 m and a minimum-points threshold of 15. Points not assigned to any dense region were retained as noise rather than allocated to a nearest cluster, preserving the distinction between genuine spatial market concentrations and dispersed individual activity. Resulting segments were profiled by customer count, mean order value, mean order frequency and aggregate expenditure. For spatial mapping, the same customer locations were additionally clustered using k-means ($k = 5$) in Google Earth Engine, with the resulting segments (Figure 4) labelled by indicative value tier consistent with the profile in Table 2.

Module 3: service geography and route sequencing

Delivery service geography was represented by aggregating ward boundaries into six compass-oriented administrative zones (North, North West, Central, East, South West and South) surrounding the reference location, produced in ArcGIS Pro from the ward layer used throughout the study (Figure 7). A delivery sequence was then computed across the depot and a set of destination points sampled from the Module 2 customer records. Route construction employed a nearest-neighbour heuristic, in which the tour is built by iteratively proceeding to the closest unvisited destination, followed by 2opt local search improvement, in which route segments are reversed wherever such reversal reduces total tour cost. This combination provides a computationally tractable approximation to a problem that is formally NPhard, and is appropriate for the destination counts demonstrated here.

Results and Discussion

Site suitability surface

The composite suitability surface, rendered as a continuous raster across Delhi, is presented in Figure 2. Scores were classified into five bands ranging from Very Low (0.20-0.31) to Very High (0.66-0.77).

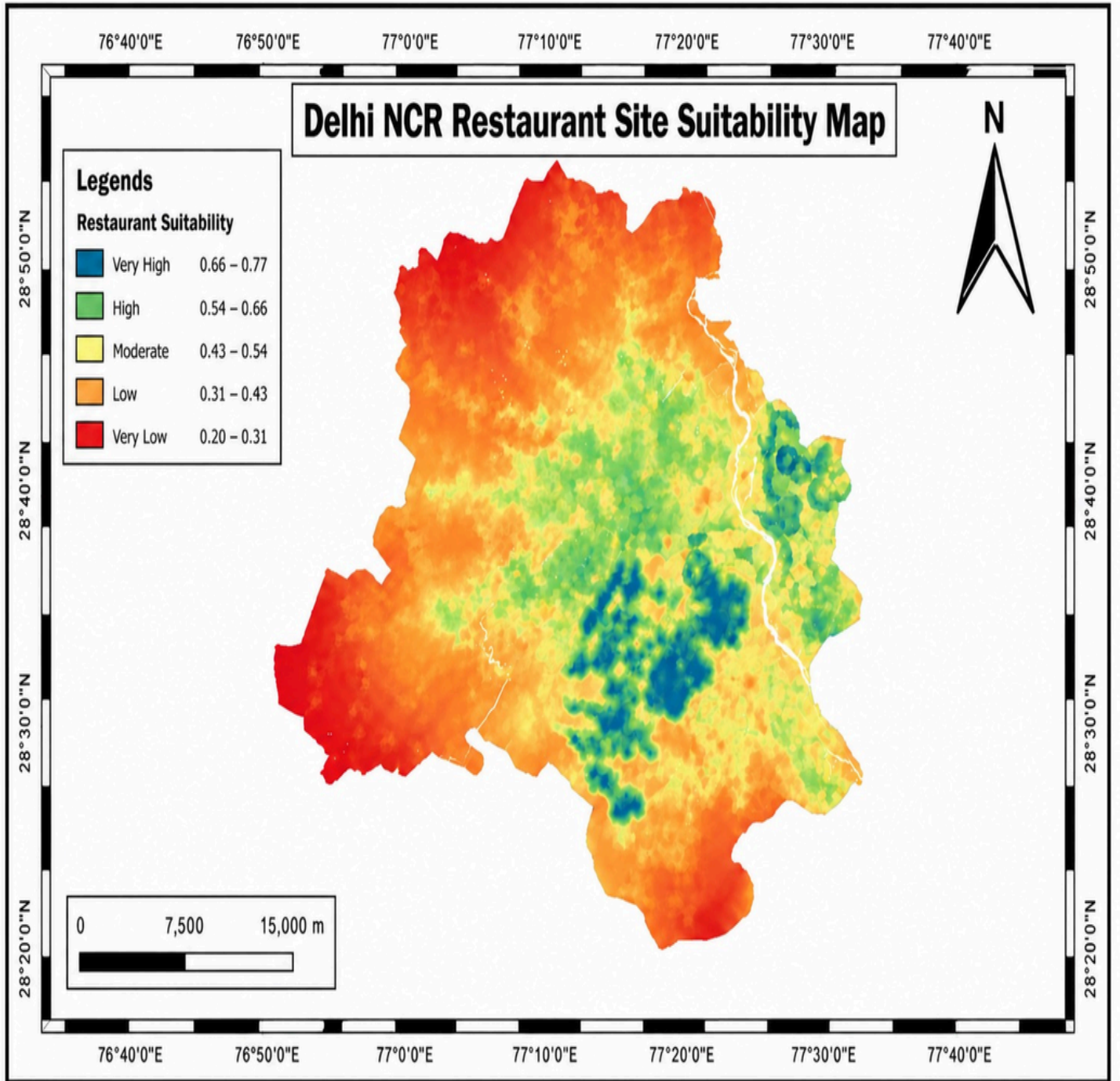


Figure 2. Restaurant site-suitability surface for Delhi. Green indicates relatively higher composite suitability; red indicates lower suitability.

The spatial structure of the surface is analytically significant. Higher-scoring locations do not form a single contiguous favourable zone but instead appear as discontinuous corridors and discrete nodes, with a prominent concentration in central and south-central Delhi and a secondary cluster near the eastern

boundary. Northern and south-western peripheral areas record consistently lower values. This granularity reflects the underlying criteria: road density, amenity proximity and competitor presence all vary substantially over distances of a few hundred metres, and the model preserves rather than smooths that variation. The practical implication is that site screening in Delhi cannot be conducted meaningfully at district resolution; discrimination is required at the level of individual street segments.

This finding is consistent with the multi-criteria logic articulated by Roig-Tierno et al. (2013), in which competing criteria favour different locations and the composite surface consequently exhibits localised optima rather than broad favourable regions.

Candidate ranking

The twenty highest-scoring candidates are summarised in Figure 3, with detailed attributes for the leading candidates given in Table 1.

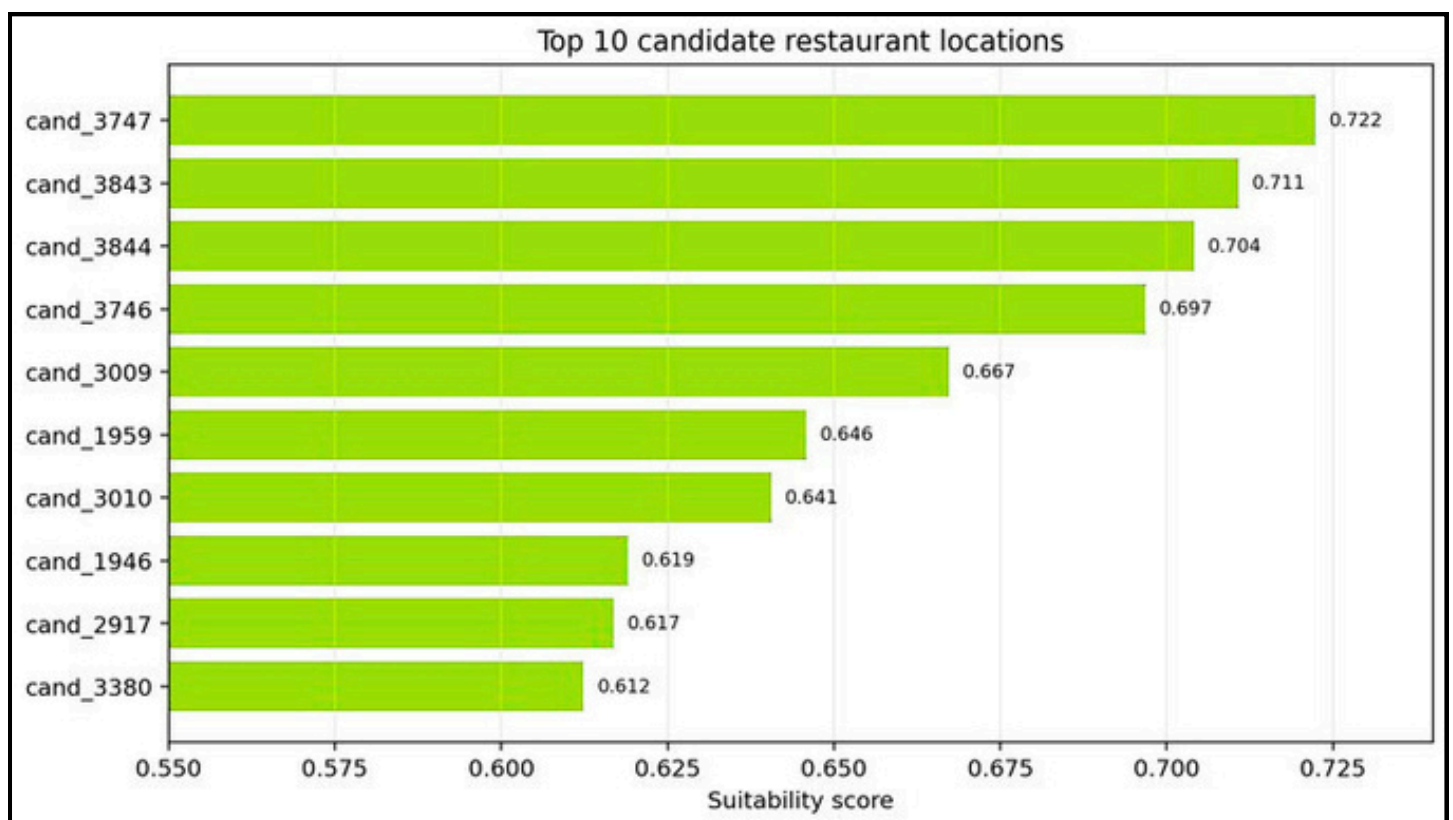


Figure 3. Top ten candidate locations ranked by composite suitability score.

Table 1. Attributes of the five highest-ranked candidate locations. Competitor, mall and metro counts are measured within a 500 m analysis neighbourhood.

Candidate	Score	Main road	Competitor	Malls	Metro
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		(m)	s		
cand_3747	0.7224	34.3	1	2	2
cand_3843	0.7108	131.9	1	2	2
cand_3844	0.7042	310.8	1	2	2
cand_3746	0.6968	93.7	1	2	2
cand_3009	0.6673	47.9	3	1	2

The leading candidate combined close main-road proximity with substantial transit and commercial amenity provision and limited mapped competition. The mean score across the top twenty was 0.6304 with a standard deviation of 0.0451, indicating a relatively compressed leading group rather than a single dominant location — an important qualification, since it implies that the ranking should be treated as identifying a shortlist warranting field investigation rather than as isolating a uniquely optimal site.

Examination of the shortlist reveals interpretable trade-offs. One candidate exhibited exceptional road proximity but comparatively weak amenity indicators; another recorded twelve competitors within 500 m, simultaneously signalling strong demonstrated demand and elevated competitive pressure; a third recorded no mapped competitors yet ranked only seventeenth, confirming that low competition alone is insufficient to generate a high composite score. This behaviour is consistent with the intended model design, in which no single criterion dominates.

A material data limitation emerged during this analysis. Office establishments were effectively absent from the retrieved point-of-interest data, with only a single office-tagged record across the entire study area, and educational institutions appeared within the analysis neighbourhood for only two shortlisted candidates. These zero values reflect the completeness variability in volunteered geographic information documented by Haklay (2010) and

confirmed for points of interest specifically by Klinkhardt et al. (2023). The affected criteria are therefore non-functional in the present implementation, and the reported scores should be understood as driven principally by road, transit, mall and competitor evidence.

Customer segmentation and market structure

DBSCAN clustering resolved five spatially distinct customer segments, with 594 customers — approximately 12 per cent of the total — retained as unclustered noise; the value-based profile of these segments is summarised in Table 2. For spatial presentation, the same customer locations were additionally clustered using k-means ($k = 5$), shown in Figure 4 with segments labelled by indicative value tier (Average Value, Budget Segment, Premium Segment, Low Mid Value and Core Segment); mapped segment sizes ranged from approximately 2,100 to 5,800 orders. A complementary continuous view of order density is provided in Figure 9.

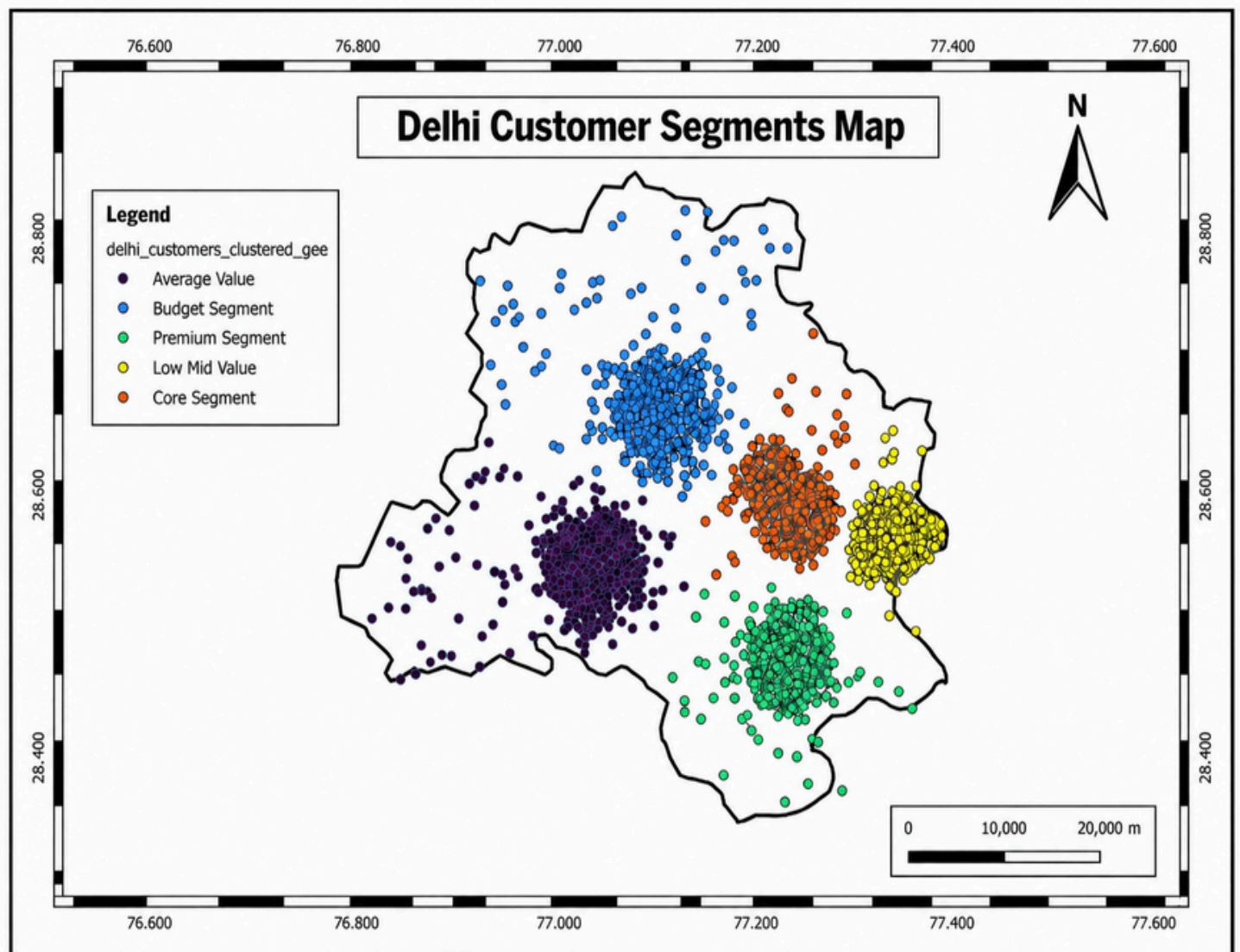


Figure 4. Spatial customer segments by indicative value tier, *k*-means clustering (*k* = 5) of synthetic customer locations.

The retention of a substantial noise fraction under DBSCAN is a methodologically meaningful result rather than a shortcoming. It confirms that the algorithm discriminated between genuine density concentrations and dispersed activity, in accordance with the design intent described by Ester et al. (1996). A partitional method such as *k*-means necessarily assigns every point to a cluster, which is why the *k*-means rendering in Figure 4 is presented as a spatial visualisation of value tier rather than as a replacement for the DBSCAN-derived profile in Table 2.

Segment characteristics are summarised in Table 2.

Table 2. Profile of the five identified customer segments. Values in Indian Rupees (INR).

Segment	Customers	Mean order value	Orders per customer	Total expenditure
0	1,412	580.11	4.01	3,295,386
1	867	749.91	3.91	2,548,516
2	738	418.42	4.01	1,242,438
3	648	379.63	4.10	1,004,878
4	513	398.87	3.99	821,642

A three-tier structure is evident. Segment 1 records the highest mean order value at INR 749.91 despite comprising fewer customers than Segment 0, indicating a premium concentration; Segment 0 combines the largest customer base with an intermediate mean value to generate the highest aggregate expenditure; and Segments 2, 3 and 4 cluster within a comparatively narrow INR 380–420 band, representing a value-oriented tier. Order frequency was near-constant across segments at approximately four orders per customer,

indicating that segment differentiation in this scenario is driven by spend per transaction rather than by purchase frequency.

Aggregated by neighbourhood (Figure 5), South Delhi generated the largest simulated value at INR 2.644 million from 3,498 orders, followed by Connaught Place at INR 2.227 million; together these accounted for approximately 50.6 per cent of total simulated value.

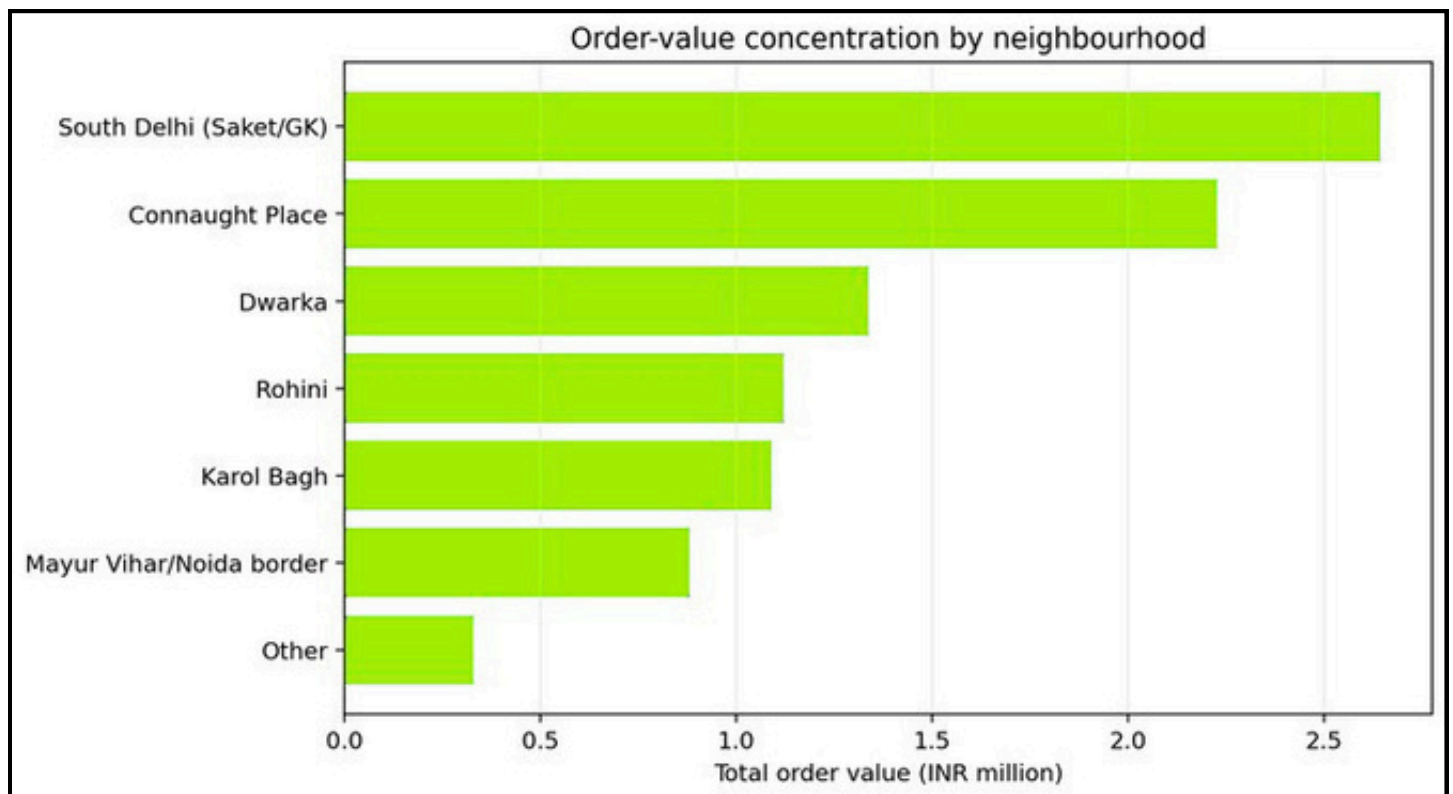


Figure 5. Order-value concentration by neighbourhood.

These patterns must be interpreted with explicit reference to their synthetic origin. The concentration observed is a property of the generative parameters used to construct the dataset and does not constitute evidence regarding actual demand distribution in Delhi. What the result does establish is that the segmentation and profiling pipeline correctly recovers spatial value structure where such structure exists, and is therefore ready to accept real transaction data without modification.

Temporal demand structure

Temporal analysis (Figure 6) identified pronounced bimodal demand concentration, with maxima at 20:00 and 21:00 recording 4,560 and 4,558 orders respectively, and a secondary midday peak at 13:00 recording 3,645 orders.

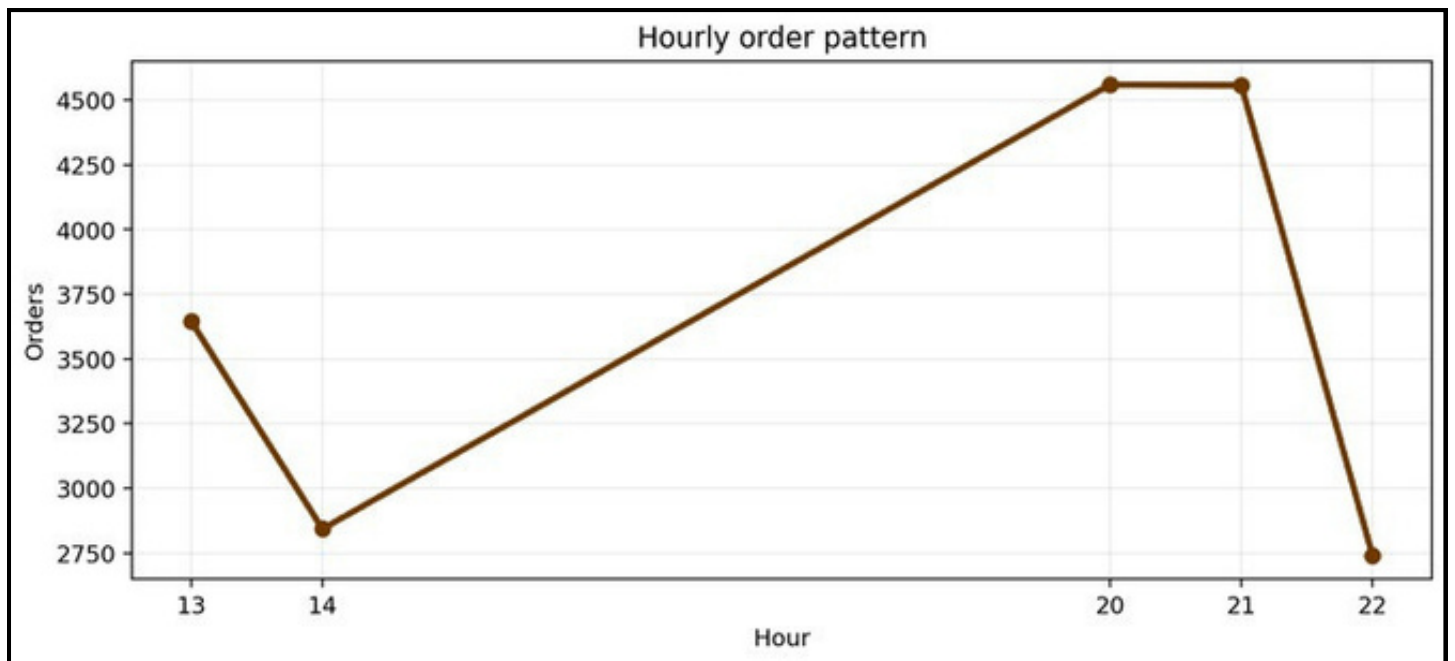


Figure 6. Hourly distribution of orders showing evening and midday demand concentration.

The operational significance lies in the implication for capacity planning. Staffing, kitchen throughput and delivery-fleet allocation must be dimensioned against peak-hour rather than mean-hour demand; provisioning to an hourly average would produce systematic service failure during the evening window that accounts for the largest share of daily volume. This finding connects Module 2 directly to Module 3, since the delivery capacity required to serve a given catchment is determined by peak concurrent load rather than by aggregate daily order count.

Service geography and delivery sequencing

Delivery service geography, aggregated from ward boundaries into six administrative zones surrounding the reference location, is shown in

Figure 7, and the optimised delivery sequence in Figure 8.

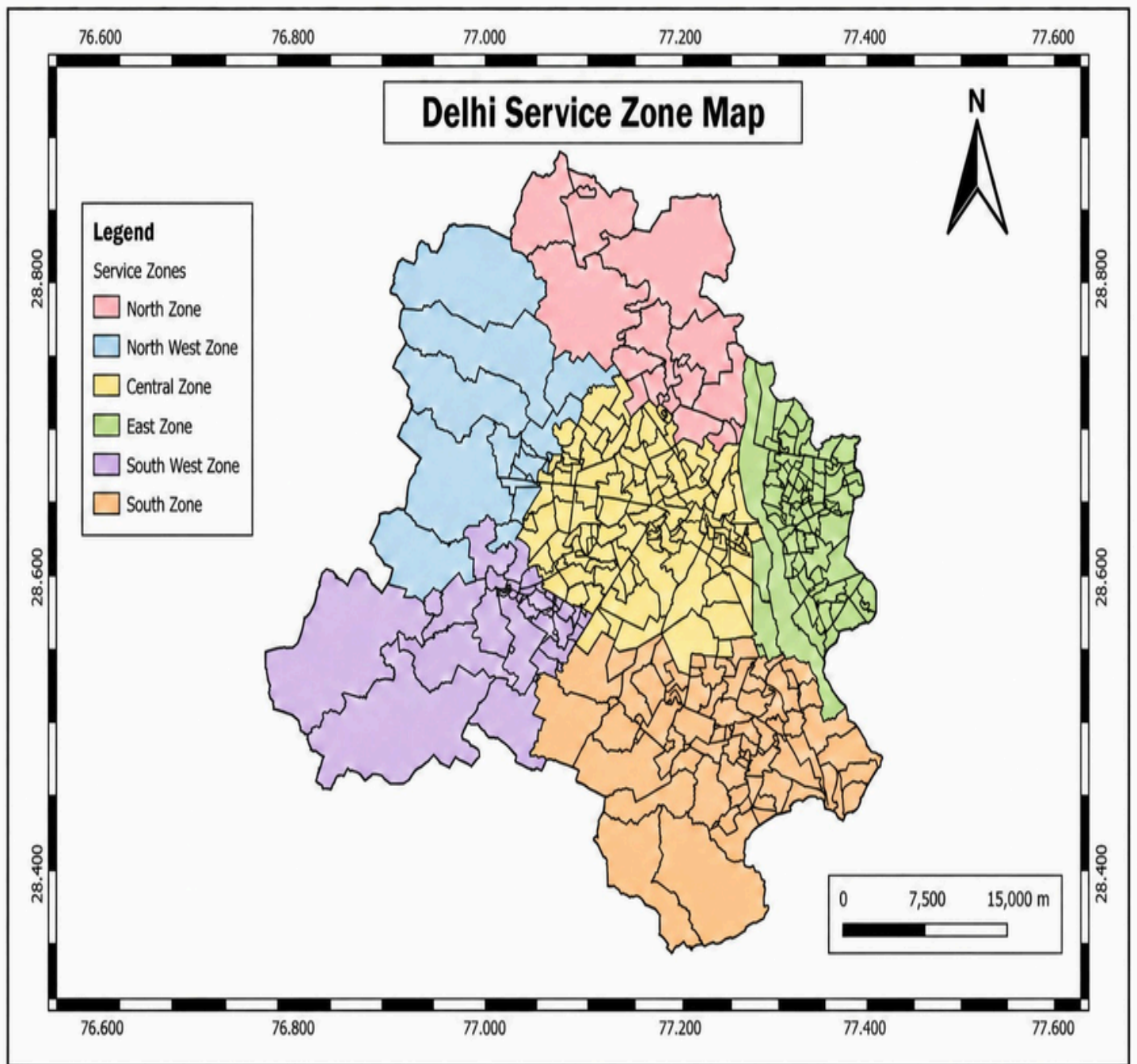


Figure 7. Delivery service zones aggregated from ward boundaries into six administrative zones surrounding the reference location.

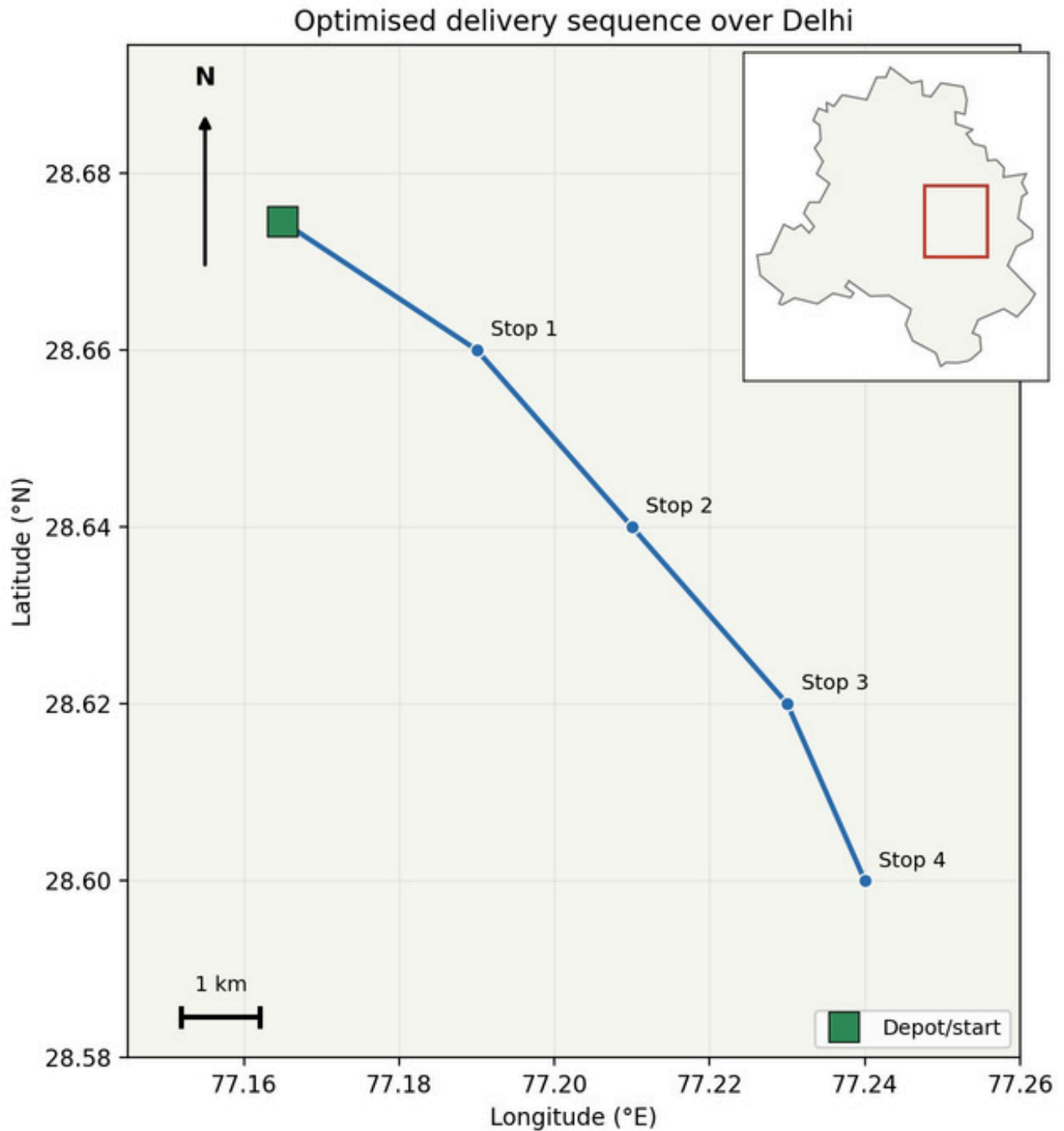


Figure 8. Optimised delivery sequence from the depot through successive destinations.

The zone structure supports differentiated service policy along genuine administrative boundaries rather than abstract radii: the Central zone, containing the reference location, may reasonably correspond to a priority rapid-delivery commitment, while the surrounding North, North West, East, South West and South zones represent progressively more peripheral catchments. The computed route progresses coherently through successive destinations without visually evident backtracking, consistent with successful 2-

opt improvement over the initial nearest-neighbour construction.

Two limitations constrain the operational interpretation of these outputs. First, administrative zone boundaries reflect ward adjacency rather than measured travel time; actual delivery reach within or across a zone will still vary with street-network structure, one-way

restrictions, physical barriers or congestion, and replacement with network travel-time isochrones computed from the OSMnx network (Boeing, 2017) constitutes a necessary refinement. Second, no quantified efficiency saving is claimed, because no independently derived baseline route and no calibrated network travel-time data were available against which to benchmark the optimised sequence. Reporting a percentage improvement absent such a baseline would convey unwarranted precision, and the result is accordingly presented as a demonstrative optimised sequence.

Customer order density is additionally shown as a continuous surface in Figure 9, complementing the discrete segment view in Figure 4.

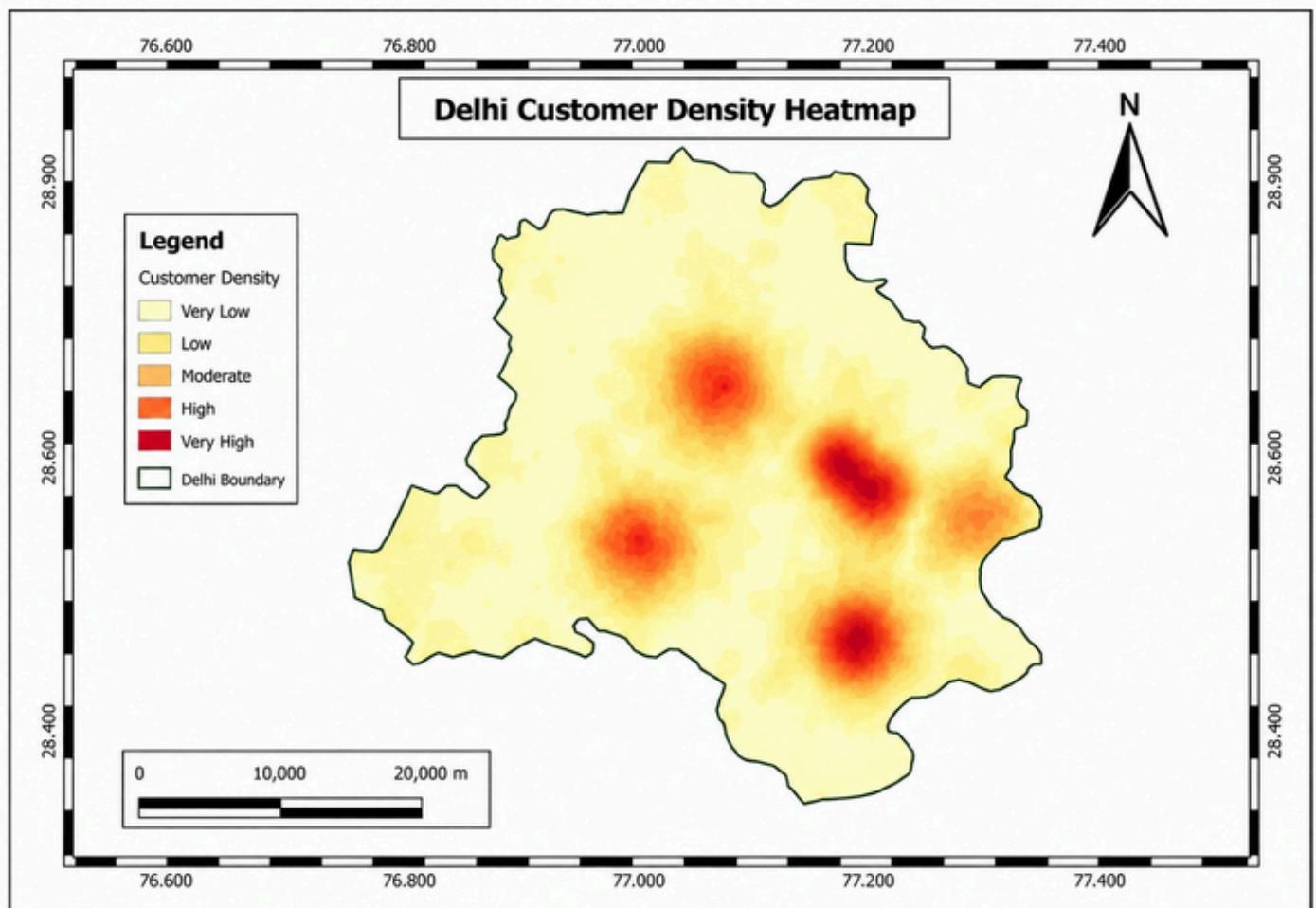


Figure 9. Continuous customer order density surface, complementing the discrete segments in Figure 4.

Network-based validation of the study extent

As a network-based validation exercise, service areas and delivery routing were additionally computed on an actual routable road network extracted for a $16.4 \text{ km} \times 17.8 \text{ km}$ sub-region of south-west Delhi (10,210 nodes, 24,471 edges, single connected component, 2,195 km of road, 4,039 one-way segments). Shortest paths were computed by Dijkstra's algorithm respecting road directionality, giving genuine network-distance service areas at 3 km, 6 km and 10 km in place of Euclidean radii. A synthetic customer distribution anchored to this road network was calibrated using DBSCAN ($\epsilon = 800 \text{ m}$, $\text{min_samples} = 15$) to reproduce the five-segment structure reported in Table 2, providing a consistency check that the segmentation approach transfers to network-constrained demand. A delivery tour across the resulting sample was constructed using nearest-neighbour initialisation followed by 2-opt and Or-opt local search on true network distances.

Because the road network is directed, the inter-stop distance matrix is asymmetric. The standard 2-opt move evaluates candidate improvements using a four-edge cost delta that assumes symmetric distances; on a directed network this assumption is invalid, since reversing a route segment also reverses the direction — and therefore the cost — of every internal edge within it. An initial implementation using the standard delta produced non-monotonic tour-length reduction, with length increasing on 9 of 27 accepted moves; recomputing the full reversed-tour cost at each candidate move corrected this, yielding provably monotonic improvement. This is a general caution for nearest-neighbour/2-opt implementations on directed street networks: the symmetric-cost assumption embedded in the standard 2-opt delta does not hold and must be addressed explicitly followed by 2-opt and Or-opt local search on true network distances. The analysis was implemented as an interactive tool permitting depot repositioning with live recomputation of service areas and routing, in addition to the static outputs reported here.

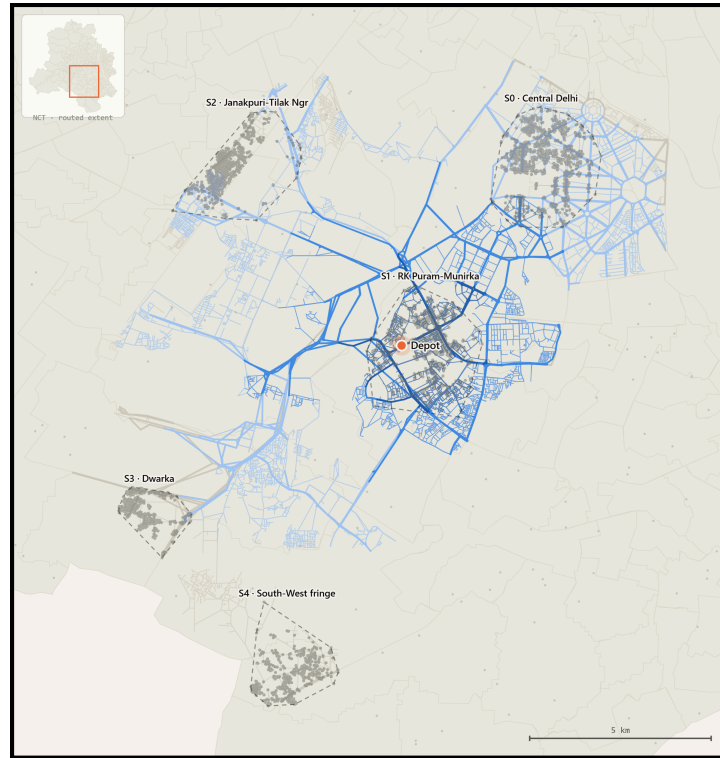


Figure 10

Figure 10 shows the road network extract together with the five customer segments, labelled by locality (S0, Central Delhi; S1, RK Puram–Munirka; S2, Janakpuri–Tilak Nagar; S3, Dwarka; S4, South-West fringe). The extract spans a contiguous diagonal band from Dwarka and Vasant Kunj through RK Puram to Connaught Place; it does not cover the full National Capital Territory, and localities such as Saket fall outside it. The depot location shown was selected, within this extract, for high road density and multi-segment reach — the qualities Module 1's suitability score rewards — rather than taken from Table 1's top-ranked candidate, whose coordinates fall outside the available road extract.

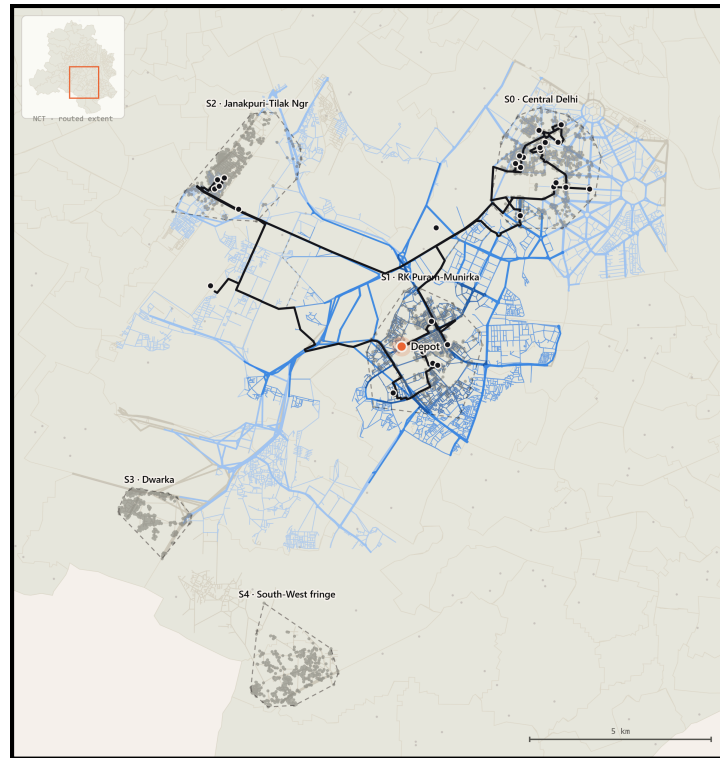


Figure 11

Figure 11 shows the optimised delivery tour across 28 stops sampled from three segments (S1, S2 and S0). Nearest-neighbour construction produced an initial tour of 23.92 km; 2-opt and Or-opt improvement reduced this to 21.06 km — a 12.0% reduction over 27 accepted moves, decreasing monotonically throughout. Across the full network, computed distances exceeded straight-line distance by a median factor of $\times 1.38$, rising to approximately $\times 1.5$ along the routed tour itself, directly quantifying the Euclidean-approximation limitation conceded in Section 4.5.

Integrated interpretation

The three modules address complementary questions at different decision horizons, and their principal analytical value emerges from their combination. Module 1 identifies where the urban context is favourable; Module 2 establishes where demand concentrates and what it is worth; Module 3 determines whether a given location can serve that demand within practical operational limits. A high suitability score does not by itself imply commercial viability: the site must additionally intersect target segments, remain affordable, and support acceptable delivery performance during peak demand. Equally, a valuable customer concentration is operationally meaningful only where it can be served reliably.

This integration is realised concretely rather than rhetorically. The reference location for Module 3 is the highest-ranked Module 1 candidate, and the delivery destinations are drawn from the Module 2 customer records. The

framework therefore produces not three parallel analyses but a single connected chain of evidence extending from strategic site screening through tactical market characterisation to operational delivery planning.

Conclusion

This study developed and demonstrated an integrated three-module GeoAI framework linking restaurant site suitability, spatial customer segmentation and delivery planning for the National Capital Territory of Delhi. The suitability model evaluated 5,946 candidate locations against six openly derived spatial criteria, producing a composite surface, classified from Very Low (0.20-0.31) to Very High (0.66-0.77), in which favourable locations formed discontinuous corridors and nodes rather than contiguous zones. Density-based clustering resolved five spatially distinct customer segments exhibiting a three-tier value structure while correctly retaining 12 per cent of customers as dispersed noise. Service-catchment definition and route sequencing translated the highest-ranked location into operational delivery geography.

The central conclusion is methodological rather than commercial. The framework establishes that strategic, tactical and operational location decisions can be connected within a single transparent and reproducible workflow, with every criterion and score open to inspection. It does not, and cannot on the present evidence base, constitute an investment recommendation: the transaction layer is synthetic, the suitability weights were reasoned rather than formally elicited, point-of-interest completeness was not independently audited, and service catchments remain ward-adjacency approximations rather than measured travel time. These constraints are stated explicitly because the distinction between demonstrated method and validated finding is fundamental to the appropriate use of such a system.

Several directions for future research follow directly. The most immediate is substitution of the synthetic transaction layer with anonymised real order records, which the pipeline is designed to accept without structural modification. Suitability weights should be re-derived through documented analytic hierarchy process elicitation and subjected to sensitivity analysis to establish the stability of the resulting ranking. Point-of-interest inputs require validation against field or commercial sources, particularly for the office and education categories found to be materially incomplete. Administrative service zones should be supplemented with network travel-time isochrones computed separately for midday and evening peak conditions, and the routing component extended to a capacitated vehicle-routing formulation incorporating delivery time windows and benchmarked against an explicit baseline. Finally, the framework's architecture is not restaurant-specific, and its extension to other location-sensitive service sectors and to additional Indian metropolitan areas would test the generality of the integration approach proposed here.

Acknowledgement

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